III. Institution of Financial System (cont.)

1. Credit cycles and financial crises

a. Credit-market sentiment and the business cycle

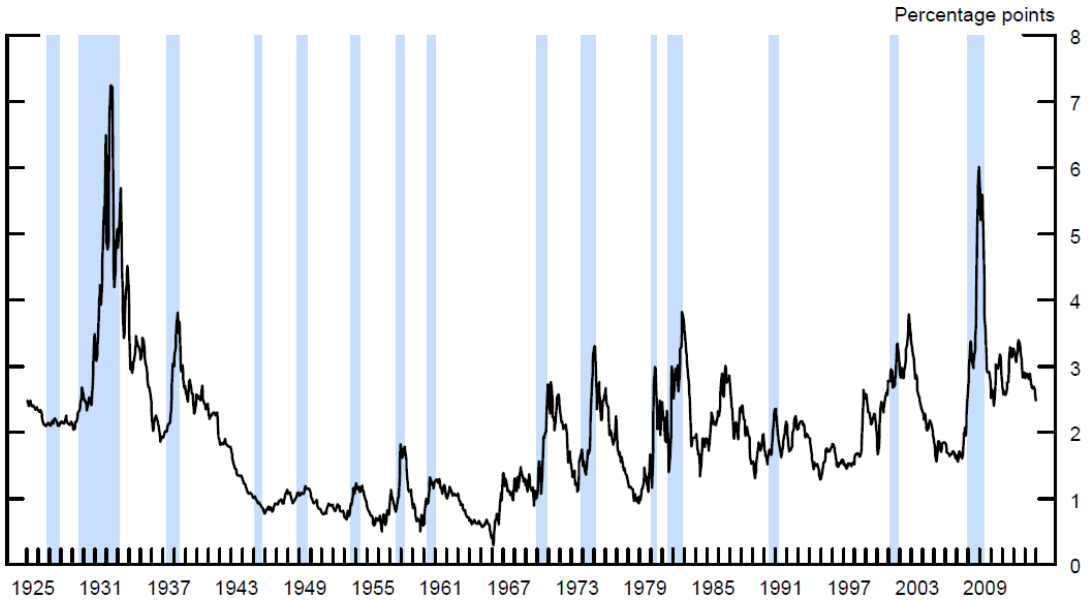
Liu, Wang, and Yang (2020) entitled "Delayed crises and slow recoveries":

"According to Minsky, the financial system is intrinsically fragile. In prosperous times, increased profits encourage the tendency toward excessive indebtedness, and sooner or later debt will accumulate to a level that borrowers can no longer pay it off with their incoming revenues and then a financial crisis follows naturally" (from Liu, Wang, and Yang (2020)).

López-Salido, Stein, and Zakrajšek (2017) entitled "Credit-Market Sentiment and the Business Cycle":

Long-time prosperity makes investors including banks tend to be over-optimistic and neglect potential risk; banks gradually take more and more risk by making loans even to borrowers with high default risk. In the end, a small default event can trigger the reversal of investors' sentiments; the sudden pullback of loans by the investors makes a financial crisis happen.

Figure 1: Baa-Treasury Credit Spread



NOTE: Sample period: monthly data from 1925:M1 to 2013:M12. The solid line depicts the spread between the yield on the Moody's seasoned Baa-rated industrial bonds and the 10-year Treasury yield. The shaded vertical bars denote the NBER-dated recessions.

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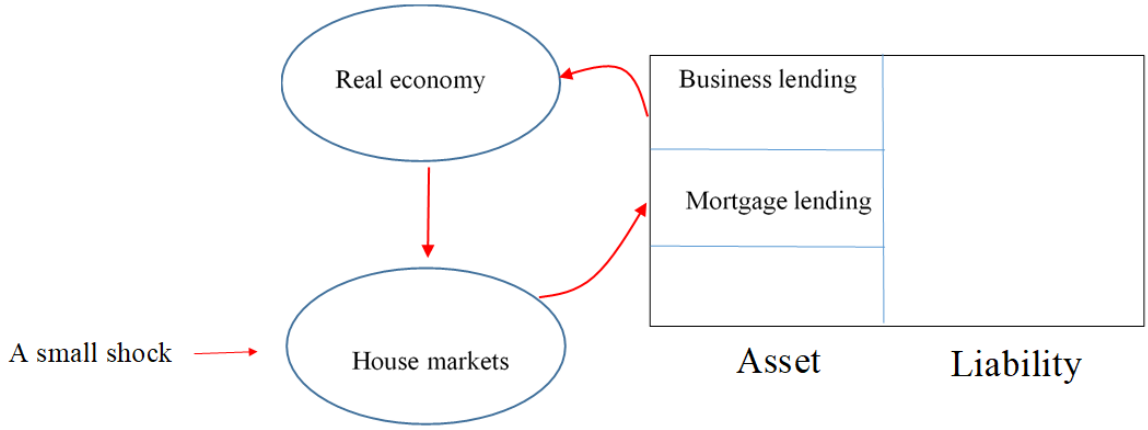
b. Procyclical lending (雨天收伞)

In good times, banks are eager to give lending; in bad times when credit is most needed, banks tend to cut lending.



Vicious cycle: house prices go down, making banks' balance sheets constrained, leading to banks contracting/reducing business lending, increasing unemployment, resulting in more property foreclosures, decreasing house prices further, and so on.

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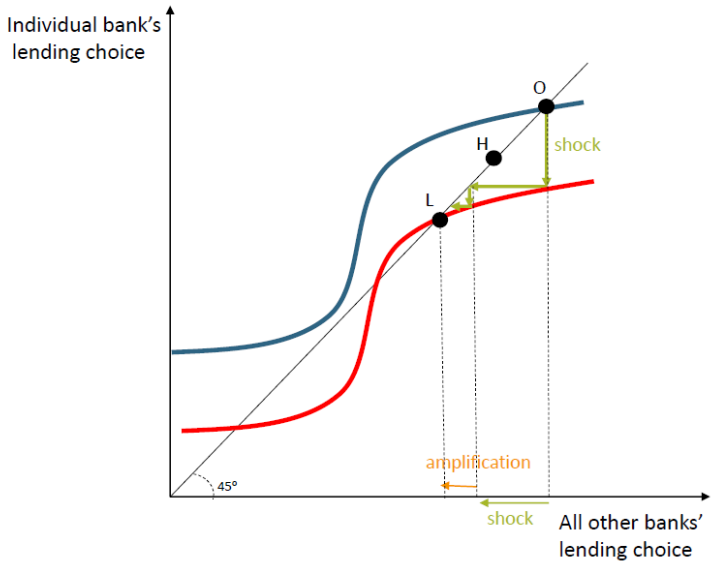
c. Strategic complementarities between banks

Brunnermeier and Reis (2016): "A Crash Course on the Euro Crisis")
see: https://scholar.princeton.edu/markus/publications/crash-course-euro-crisis

A bank's optimal action = f (fundamental, other banks' actions)

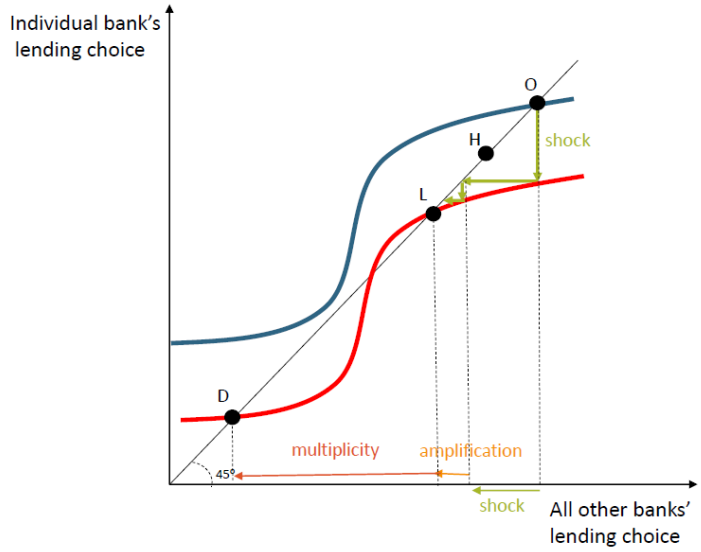
Amplification and liquidity spirals

- Now when one bank cuts lending, the others want to cut lending further.
- Leads to further cuts, and so on
- Since all the banks' actions are now strategic complements, it leads to amplification of the initial shock
- Equilibrium is now lower at *L*



Multiple equilibrium

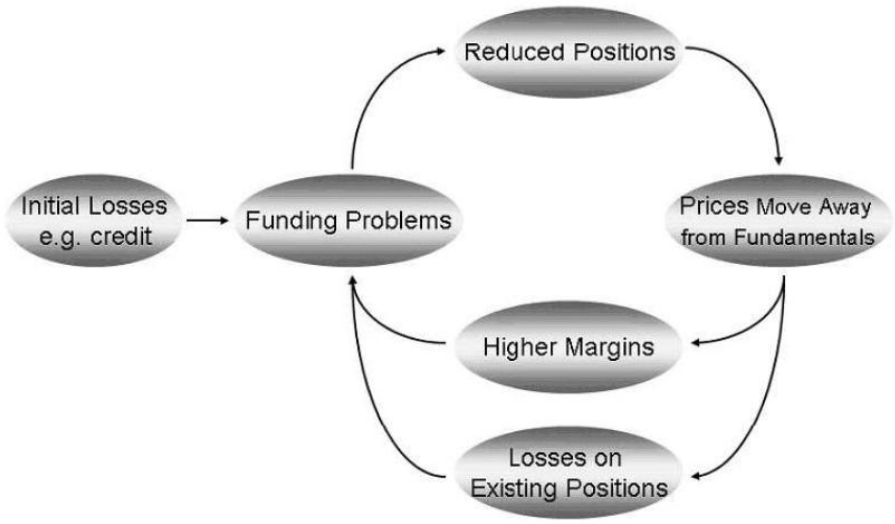
- Shifts in beliefs can lead to a new (stable) equilibrium at a much lower point *D*
- If banks believe that others will cut lending, then they anticipate fire sales and price drops, and the losses and margins spirals
- They will cut lending beforehand due to anticipation and thus triggering the depressed-lending equilibrium
- Hence, there is multiplicity of equilibrium



(Source: Brunnermeier and Reis (2016) "A Crash Course on the Euro Crisis")

d. Amplification mechanism: the two liquidity spirals -- loss spiral and margin spiral

Brunnermeier (2010) entitled "Deciphering the Liquidity and Credit Crunch 2007-08"
See: https://scholar.princeton.edu/markus/publications/deciphering-liquidity-and-credit-crunch-2007-2008

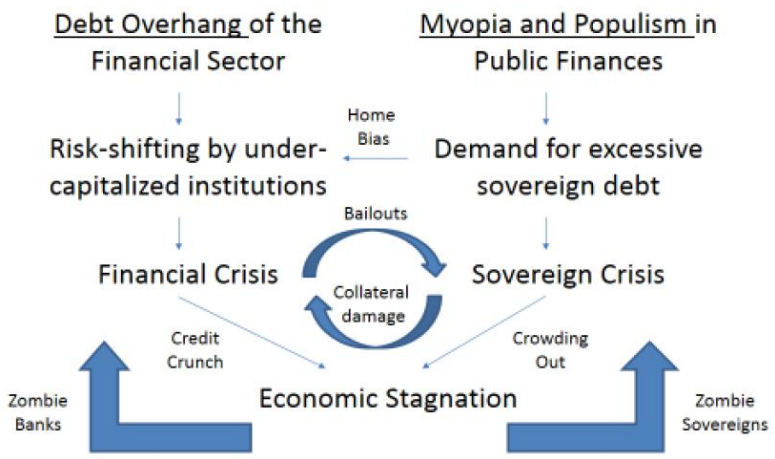


Note: Funding problems force leveraged investors to unwind their positions causing 1) more losses and 2) higher margins and haircuts, which in turn exacerbates the funding problems and so on.
Source: Brunnermeier and Pedersen (2009).

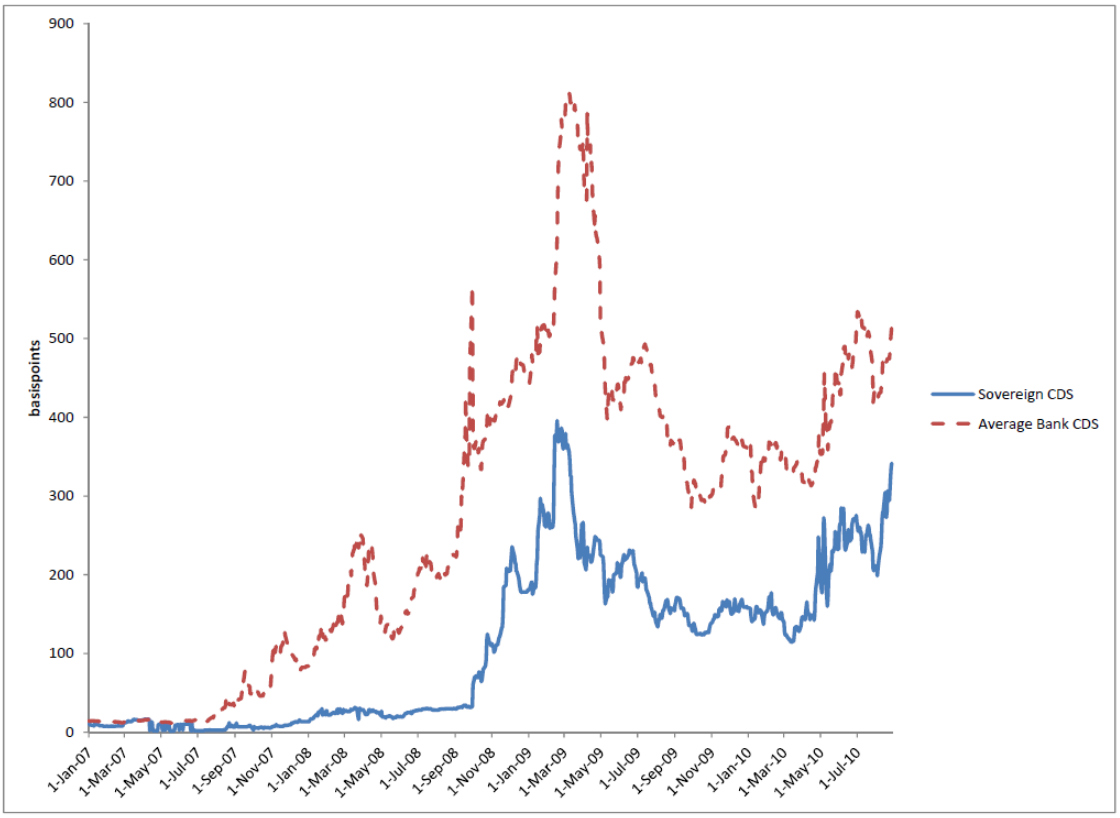
2. Twin crises

Financial liberalization leads to excessive borrowing from foreign countries. A shock (e.g., a sudden increase in interest rates in foreign countries like USA) can trigger **twin crises** that are simultaneous crises in banking and currency.

(The eurozone crisis also featured **twin crises**: banking crisis and sovereign debt crisis.)



From Acharya (2020)



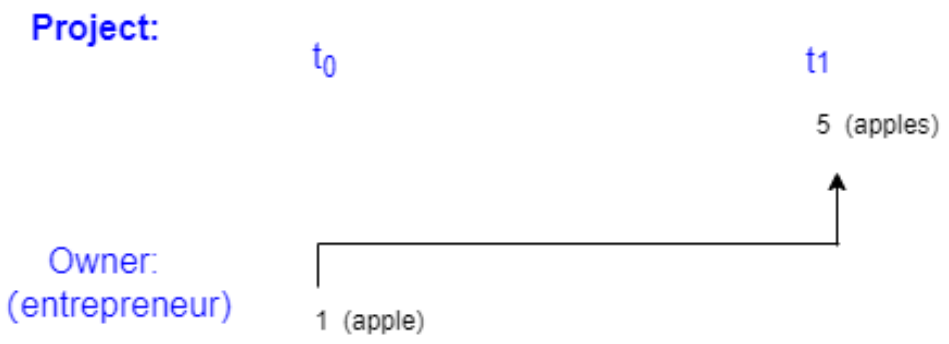
Source: Acharya et al. (2014)

3. Harm of inflation

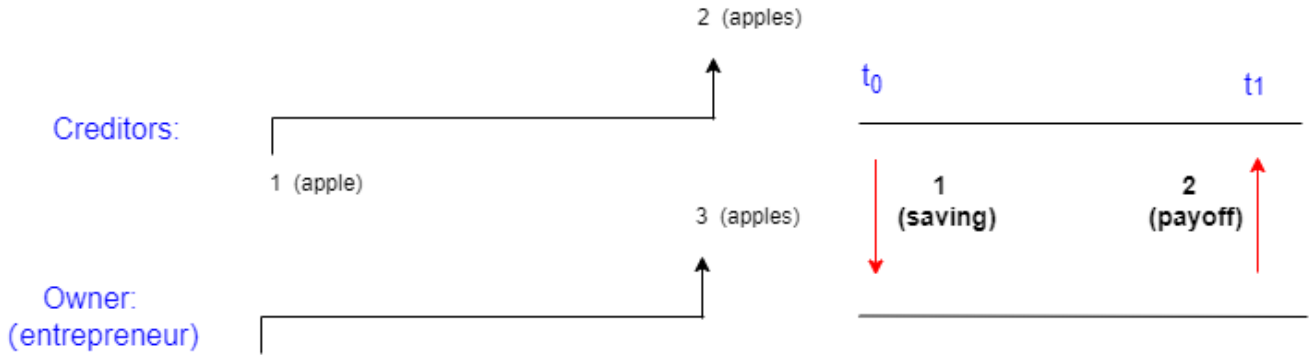
i) Price changes are not symmetric across different asset classes, so income inequality arises, i.e., some people (say, holding real estates and stocks, or with more financial knowledge) benefit more, while some other people (say, holding cash) suffer.

ii) Dynamically, inflation rates can be volatile and uncertain, which discourages savings and consequently leads to lower aggregate investment and higher unemployment.

Illustration:



Contract between two parties:



Real (gross) interest rate is 200%

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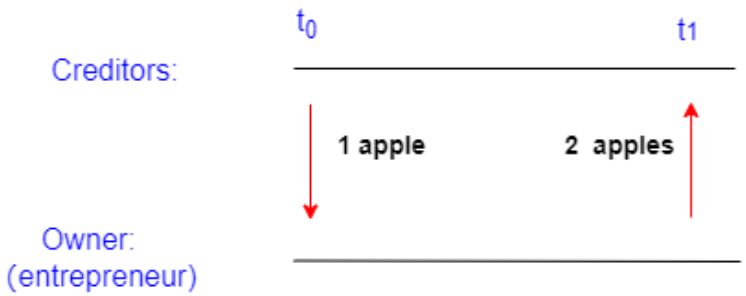
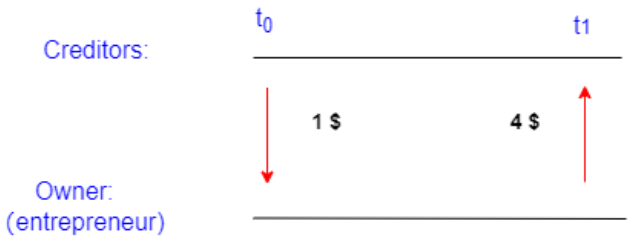
Pay by real goods:



Pay by fiat money:

Suppose inflation rate is **certain/perfectly foreseen**

$ 1 per apple at t_0; $ 2 per apple at t_1; inflation rate i is 200%

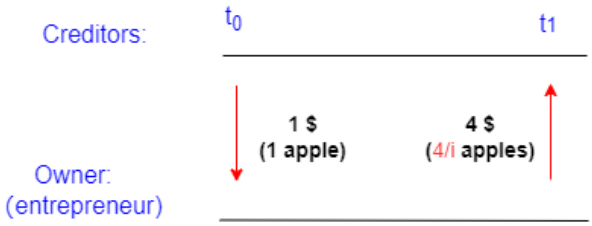


Real economy is not affected!

Pay by fiat money:

Suppose inflation rate is with **uncertainty**

inflation rate i is a random variable



Real economy is affected!

Facing high uncertainty of inflation rate, households would save less and consume more, so aggregate investment of firms goes down, causing higher unemployment.

Essentially, in this case, money loses its function of storing value.

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4. Federal funds and QEs

Definition: **The market for federal funds** is a market for *unsecured* loans of reserve balances at the Federal Reserve Banks, that allows participants with excess reserve balances to lend (or sell funds) to those with reserve balance **shortages**. https://www.newyorkfed.org/medialibrary/media/research/staff_reports/sr549.pdf

i) When can shortage happen?

Individual bank A can be short of reserves for three reasons. 1) Official reserve requirement; 2) some depositors of banks A want to withdraw cash only, and do not accept bank A's IOU/liability/electronic transfer; 3) Some counterparty creditor banks of Bank A do not want to accept Bank's IOU/liability/electronic transfer.

For the 2) and 3) reasons, Bank A has to use reserves (deposit in the central bank) to clear the financial transactions. So even without official reserve requirement –reason 1), banks may still need some reserves.

ii) However, when the central bank sets the effective reserve fund rate as zero, individual banks effectively can borrow any amount of reserves in the interbank market with zero cost. So even if individual banks need to borrow reserves for the purposes of 2) and 3) above, those banks can easily do that with zero cost.

Therefore, when the effective reserve fund rate is zero, theoretically individual banks can expand their balance sheets infinitely if they want. This is unlike the case with positive reserve requirement and a positive reserve fund rate, where banks face hard constraints in expanding balance sheet.

iii) Why individual banks do not expand their balance sheet infinity although they can?

First, in the new regime after 2008, there are two implicit costs for expanding balance sheet. A) there are FDIC insurance fees when banks increase their asset/liability (deposits); B) Financial regulation of liquidity requirements have created incentives for banks to maintain substantial buffers of reserves (and other high-quality liquid assets).
https://www.newyorkfed.org/medialibrary/media/research/staff_reports/sr840.pdf

Second, although individual bank can borrow with zero funding cost to invest/lend, their equityholders bear the first loss before their creditors. So the profit-maximizing banks will trade off the risk-return in their lending.

iv) How the central banks can implement the monetary policy (i.e., lower interest rates) in the new regime where the effective reserve fund rate is already zero?

Indeed, after the effective reserve fund rate is zero, the interbank trading among banks become rare. So the central bank cannot influence the reserve fund rate any longer.

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https://libertystreeteconomics.newyorkfed.org/2013/12/whos-borrowing-in-the-fed-funds-market.html
http://libertystreeteconomics.newyorkfed.org/2013/12/whos-lending-in-the-fed-funds-market.html

In this case, the central bank relies on buying long-term assets, so to directly influence the long-term interest rates. This is QEs.

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